

Assignment 2 : solns.

1. A particle of mass m is confined in a one-dimensional box of length L . The state of the particle is given by the normalized wavefunction $\psi(x) = \frac{1}{3}\psi_1(x) + \frac{i}{3}\psi_3(x) - \sqrt{\frac{7}{9}}\psi_5(x)$, where $\psi_n(x)$ is a normalized particle-in-a-box wavefunction corresponding to quantum number n . (a) What will be the outcome when the energy of the particle is measured? (b) If more than one result is possible, give the probability of obtaining each result. (c) What is the expectation value of the energy? (d) If the linear momentum of the particle described above were measured, what would we expect to find?

Ans. A single measurement of the energy yields a single outcome which is one of the eigenvalues E_n (associated with the eigenfunction n appearing in the expansion of $\psi(x)$). The probability of obtaining E_n is c_n^2 where c_n = coefficient of n in the expansion.

(a) When the energy of the particle is measured, possible outcomes are

$$E_1 = \frac{h^2}{8mL^2}, E_3 = \frac{9h^2}{8mL^2}, E_5 = \frac{25h^2}{8mL^2}$$

(b) The probability of obtaining each result is $|c_1|^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$, $|c_3|^2 = \left|\frac{i}{3}\right|^2 = \frac{1}{9}$, $|c_5|^2 = \frac{7}{9}$

(c) The expectation value is the weighted sum of the possible eigenvalues: $\frac{1}{9}E_1 + \frac{1}{9}E_3 + \frac{7}{9}E_5 = \frac{185h^2}{72mL^2}$

(d) It is a free particle in the box, so the average momentum will always be zero. We can confirm it by calculating the momentum for each eigenstate as $\langle p \rangle_n = \int \psi_n^* \left(\frac{\hbar}{i} \frac{d}{dx} \right) \psi_n dx = 0$

2. Consider vector spaces, $\mathcal{V}$, $\mathcal{W}$ and $\mathcal{X}$ and linear operators $\hat{A}$ and $\hat{B}$:

$\hat{A} \{ \mathcal{V} \} \mapsto \{ \mathcal{W} \}$ and $\hat{B} \{ \mathcal{W} \} \mapsto \{ \mathcal{X} \}$.

Show that the matrix representation for the linear transformation $\hat{B}\hat{A}$ is the matrix product of the matrix representations for $\hat{B}$ and $\hat{A}$ with respect to appropriate bases.

Ans. Let us say that the dimensionality of the vector spaces, $\mathcal{V}$, $\mathcal{W}$ and $\mathcal{X}$ are m , n and p respectively.

then the matrices corresponding to operators $\hat{A}$ and $\hat{B}$, say $\mathbf{A}$ and $\mathbf{B}$ respectively, must be of dimensions $n \times m$, and $p \times n$ respectively.

The linear transformation $(\hat{B}\hat{A}) \{ \mathcal{V} \} \mapsto \{ \mathcal{X} \}$ corresponds to a matrix of dimension $p \times m$

consider vectors $|V\rangle \in \mathcal{V}$ s.t. $\hat{A}|V\rangle = |W\rangle \in \mathcal{W}$ and $\hat{B}|W\rangle = |X\rangle \in \mathcal{X}$

The corresponding matrix representations will be $\mathbf{A}_{n \times m} \mathbf{V}_{m \times 1} = \mathbf{W}_{n \times 1}$ and $\mathbf{B}_{p \times n} \mathbf{W}_{n \times 1} = \mathbf{X}_{p \times 1}$

$$\implies \mathbf{B}_{p \times n} (\mathbf{A}_{n \times m} \mathbf{V}_{m \times 1}) = \mathbf{X}_{p \times 1} \text{ or, } (\mathbf{B}_{p \times n} \mathbf{A}_{n \times m}) \mathbf{V}_{m \times 1} = \mathbf{X}_{p \times 1}$$

The product $(\mathbf{B}_{p \times n} \mathbf{A}_{n \times m})$ is a new matrix of dimension $p \times m$.

3. Evaluate the commutators (a) $[\hat{x}, \hat{p}_x^2]$ (b) $[\hat{x}, \hat{H}]$

Ans. (a) First prove that for operators $\hat{A}, \hat{B}, \hat{C}$, $[A, BC] = [A, B]C + B[A, C]$

Operating on a function f , LHS = $[A, BC]f = [ABC - BCA]f$

$$\text{RHS} = [A, B]Cf + B[A, C]f = [AB - BA]Cf + B[AC - CA]f = ABCf - BACf + BACf - BCAf = [ABC - BCA]f$$

Hence QED.

$$\therefore [\hat{x}, \hat{p}_x^2] = [\hat{x}, \hat{p}_x]\hat{p}_x + \hat{p}_x[\hat{x}, \hat{p}_x] = 2i\hbar\hat{p}_x$$

$$(b) [\hat{x}, \hat{H}] = [\hat{x}, \frac{\hat{p}_x^2}{2\mu} + \hat{V}(x)] = \frac{1}{2\mu} [\hat{x}, \hat{p}_x^2] + [\hat{x}, \hat{V}(x)] - \text{the second term is zero.}$$

$$= \frac{i\hbar}{\mu} \hat{p}_x$$

4. (a) The ground-state wavefunction of a hydrogenic atom has the form $\psi(r) = Ne^{-Zbr}$, where $b = \frac{1}{a_0}$ and a_0 is the Bohr radius, 52.9 pm. $Z = 1 \implies$ H atom.

Normalize this spherically symmetrical function.

$$\text{Ans. } \int_{\text{all space}} |\psi|^2 d\tau = N^2 \int_0^\infty d\phi \int_0^\pi \sin\theta d\theta \int_0^\infty r^2 e^{-2Zbr} dr = N^2 \cdot 2\pi \cdot 2 \cdot \frac{1}{4Z^3 b^3}. \text{ Equating this to 1 gives } N = \sqrt{\frac{Z^3 b^3}{\pi}} = \sqrt{\frac{Z^3}{a_0^3}} = 1.5 \times 10^{15} \text{ m}^{-\frac{3}{2}} \text{ for } Z = 1.$$

(b) A particle is confined to the region $0 \leq x \leq \infty$ and its state is described by the unnormalized wavefunction e^{-2x} .

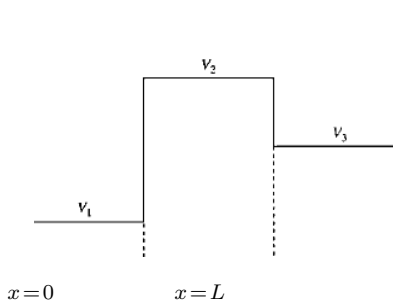
What is the probability of finding the particle at a distance $x > 1$?

$$\text{Ans. The required probability} = \frac{\int_1^\infty (e^{-2x})^2 dx}{\int_0^\infty (e^{-2x})^2 dx} = \frac{\int_1^\infty e^{-4x} dx}{\int_0^\infty e^{-4x} dx} = 1 - \frac{\int_0^1 e^{-4x} dx}{\int_0^\infty e^{-4x} dx} = 1 - \frac{e^{-4x} \Big|_0^1}{e^{-4x} \Big|_0^\infty} = 1 - \frac{e^{-4} - 1}{0 - 1} = e^{-4}$$

5. Consider the 1-D space in which a particle can experience one of three potentials depending upon its position.

They are: $V_1 = 0 \forall -\infty < x < 0$, $V = V_2$ for $\forall 0 \leq x < L$, and $V = V_3$ for $\forall L \leq x < \infty$. The particle wavefunction is to have both a component e^{ik_1x} incident upon the barrier V_2 and a reflected component in region A. In region C the wavefunction has only a forward component. The energy of the particle, E , is in the range $V_2 > E > V_3$. (a) Derive a general equation for transmission probability, T ; (b) Show that the general equation for T reduces to the one derived in the class in the high, wide barrier limit when $V_3 = 0$. (c) Draw a graph of the probability of proton tunnelling when $V_3 = 0$, $L = 50\text{pm}$, and $E = 10 \text{ kJ mol}^{-1}$ in the barrier range $E < V_2 < 2E$.

Ans.



Region 1

Region 2

Region 3

$$\psi_1(x) = Ae^{ikx} + A'e^{-ikx}; \quad k = \sqrt{\frac{2\mu E}{\hbar^2}}$$

$$\psi_2(x) = Be^{\kappa x} + B'e^{-\kappa x}; \quad \kappa = \sqrt{\frac{2\mu(V_2 - E)}{\hbar^2}}$$

$$\psi_3(x) = Ce^{ik_3x}; \quad k_3 = \sqrt{\frac{2\mu(E - V_3)}{\hbar^2}}$$

using B.C. : $\psi_1(0) = \psi_2(0)$; $\psi_1'(0) = \psi_2'(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

using $\psi_2(L) = \psi_3(L)$; $\psi'_2(L) = \psi'_3(L)$

$$\Rightarrow B e^{\kappa L} + B' e^{-\kappa L} = C e^{\alpha_3 L}; \alpha_3 = ik_3$$

and $\kappa B e^{\kappa L} - \kappa B' e^{-\kappa L} = \alpha_3 C e^{\alpha_3 L}$

then: $\eta B + \eta^{-1} B' = \beta C$ and $\kappa \eta B - \kappa \eta^{-1} B' = \alpha_3 \beta C$
write: $\eta = e^{\kappa L}$; $\beta = e^{\alpha_3 L}$

get: $2\kappa \eta B = \beta C(\kappa + \alpha_3)$ and $2\kappa \eta^{-1} B' = \beta C(\kappa - \alpha_3)$

We need C in terms of A

$$2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha_3)}{2\kappa \eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha_3)}{2\kappa \eta^{-1}} = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)(\kappa + \alpha_3)}{\eta} + \frac{(\alpha - \kappa)(\kappa - \alpha_3)}{\eta^{-1}} \right]$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)(\kappa + \alpha_3)}{\eta} + \frac{(\alpha - \kappa)(\kappa - \alpha_3)}{\eta^{-1}} \right]} \stackrel{\alpha=ik}{=} \frac{1}{\alpha_3=ik_3} \frac{1}{\beta} \frac{4ik\kappa}{-(\eta - \eta^{-1})(\kappa^2 - k k_3) + i\kappa(k + k_3)(\eta + \eta^{-1})}$$

$$\text{and } T = \left| \frac{C}{A} \right|^2 \stackrel{|\beta|^2=1}{=} \frac{16\kappa^2 \kappa^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k k_3)^2 + \kappa^2 (k + k_3)^2 (\eta + \eta^{-1})^2} = \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k k_3)^2 + \kappa^2 (k + k_3)^2 [(\eta - \eta^{-1})^2 + 4]}$$

$$= \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 [(\kappa^2 - k k_3)^2 + \kappa^2 (k + k_3)^2] + 4\kappa^2 (k + k_3)^2} = \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 + k^2) (\kappa^2 + k_3^2) + 4\kappa^2 (k + k_3)^2} = \frac{1}{\frac{(\eta - \eta^{-1})^2 (\kappa^2 + k^2) (\kappa^2 + k_3^2)}{16\kappa^2 k^2} + \frac{(k + k_3)^2}{4k^2}}$$

$$\text{or, } T = \frac{|C|^2}{|A|^2} = \frac{4k^2}{(k + k_3)^2} \frac{1}{1 + G}, \text{ where } G = \frac{(\eta - \eta^{-1})^2}{\frac{4\kappa^2 (k + k_3)^2}{(\kappa^2 + k^2)(\kappa^2 + k_3^2)}}$$

(b) For the case $V_3 = 0, k = k_3$ and $T = \frac{1}{1 + G}$, where $G = \frac{(\eta - \eta^{-1})^2}{\frac{16\kappa^2 k^2}{(\kappa^2 + k^2)^2}}$, the expression derived in the class.

(c) $V_3 = 0, L = 50 \text{ pm}$, and $E = 10 \text{ kJ mol}^{-1}$ in the barrier range $E < V_2 < 2E$.

First note that for $V_3 = 0$, we have the case we discussed in the class. So, we can use the expressions obtained for this simpler case.

Transmission prob. $T = \frac{1}{1 + G}$, where $G = \frac{(\eta - \eta^{-1})^2}{16\epsilon(1 - \epsilon)}$; $\eta = e^{\kappa L} = e^{\frac{L}{D}}$ and $D = \frac{\hbar}{\sqrt{2\mu(V - E)}} > 0 \quad \forall V > E$

The plots shown in the class were for T against ϵ .

I am pasting a plot here for T against $1/\epsilon$.

